DebateTrader

A Multi-Agent LLM Framework with Structured Bull-Bear Debate for Stock Trading

DSAN 6725 Final Project Presentation

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Problem & Motivation

Why Stock Trading Is Hard

- Market decisions combine price action, fundamentals, macro conditions, and news.
- These evidence sources often point in different directions.
- A single LLM agent can produce a plausible recommendation without fully exposing counterarguments.

Why Structured Debate

Diverse Evidence

Specialized agents analyze different financial signals.

Balanced Reasoning

Bull and Bear agents present competing interpretations.

Inspectable Decisions

The Judge agent evaluates both sides before producing a final decision.

Portfolio-Aware Output

Ticker-level decisions are compared before allocation.

Can Structured Disagreement Improve LLM-Based Trading Decisions?

Specialized Analysis

Can different agents analyze technical, fundamental, macro, and news-based evidence separately?

Structured Debate

Can Bull and Bear agents make competing interpretations explicit?

Judge-Based Decision

Can a Judge agent compare both sides and produce a more balanced ticker-level decision?

Portfolio Translation

Can ticker-level decisions be converted into portfolio weights with risk controls?

The project tests whether multi-agent debate can make LLM trading decisions more modular, interpretable, and risk-aware.

System Overview

DebateTrader converts raw financial evidence into final portfolio allocations through a modular seven-stage pipeline.

01

Data Collection

Collect price, fundamentals, macro, news, and trend data.

02

Feature Snapshots

Build standardized weekly evidence snapshots for each ticker.

03

Analyst Reports

Generate technical, fundamental, macro, and news/trend reports.

04

Bull–Bear Debate

Use competing agents to argue bullish and bearish cases.

05

Judge Decision

Produce ticker-level investment decisions from the debate.

06

Portfolio Allocation

Compare all six tickers and assign portfolio weights.

07

Risk Management & Backtest

Apply deterministic constraints and evaluate historical performance.

Each stage produces inspectable intermediate outputs, making the pipeline easier to debug and evaluate.

System Overview

DebateTrader converts raw financial evidence into final portfolio allocations through a modular seven-stage pipeline.

1

Stock Universe

6 S&P 500 stocks: AAPL, AMZN, BRK.B, GOOGL, LLY, XOM

2

Data Collection

Prices, Fundamentals, Macro Data, News, Trends

3

Feature Snapshots

Standardized weekly evidence snapshots for each ticker

4

Analyst Reports

Technical, Fundamental, Macro, News & Trend

5

Bull–Bear Debate

Bull Agent

Bear Agent

Judge Agent

6

Portfolio Allocation

Compare all six tickers and assign weights

7

Risk Management

Apply position and portfolio constraints

8

Backtesting

Evaluate historical strategy performance

DebateTrader separates data processing, analysis, reasoning, allocation, and evaluation into modular stages.

Stock Universe & Evaluation Scope

The system is evaluated on a controlled six-stock S&P 500 universe to keep the experiment diverse but inspectable. Using one stock from each sector gives the portfolio cross-sector exposure while keeping the agent workflow manageable for weekly evaluation.

AAPL
Information Technology

AMZN
Consumer Discretionary

BRK.B
Financials

GOOGL
Communication Services

LLY
Health Care

XOM
Energy

Multi-Source Financial Evidence

Data Sources

Yahoo Finance
OHLCV prices and supplementary fundamentals

SEC EDGAR
Financial statements

FRED
Macroeconomic indicators

Finnhub News API
Company news

Google Trends
Retail attention signals

Analyst Modules

- Technical Analyst
- Fundamental Analyst
- Macro Analyst
- News & Trend Analyst

Structured Outputs

- Signal
- Confidence
- Summary
- Bullish factors
- Bearish factors
- Risk flags
- Key metrics

Different evidence sources are transformed into standardized analyst reports before entering the debate stage.

Specialized Analyst Agents

Each analyst produces a standardized JSON report with signal, confidence, summary, bullish factors, bearish factors, risk flags, and key metrics.

Technical Analyst

- SMA, EMA, RSI, MACD
- Returns and volatility
- Relative volume
- Price-based momentum signals

Fundamental Analyst

- Valuation and growth
- Margins and profitability
- Leverage and liquidity
- ROE, ROA, and cash flow

Macro Analyst

- Interest rates and yield curve
- VIX and credit spreads
- GDP and unemployment
- Inflation indicators

News & Trend Analyst

- Company news volume
- Sentiment signals
- Uncertainty and litigiousness flags
- Google Trends attention signals

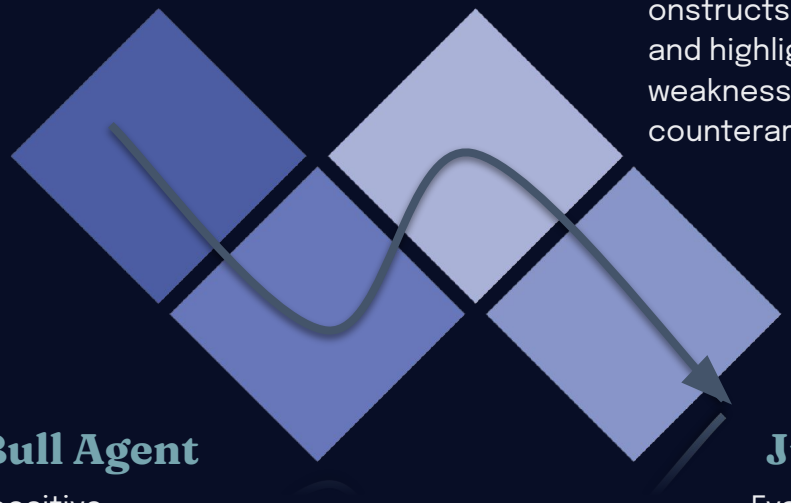
Structured Bull–Bear Debate

Analyst Reports

Technical, fundamental, macro, and news/trend evidence enters the debate stage.

Bear Agent

onstructs the negative case and highlights risks, weaknesses, and counterarguments.



Bull Agent

Constructs the positive investment case using the provided analyst evidence.

Judge Agent

Evaluates both sides and produces the final ticker-level decision.

Debate memory provides lightweight cross-week context, but cannot override current analyst evidence.

Cross-Ticker Portfolio Reasoning

Portfolio construction happens at the portfolio level, not one stock at a time.

Debate Outputs

- Collect final outputs from the ticker-level debate stage
- Include the Judge's decision for each stock
- Preserve the main bullish and bearish arguments
- Summarize stock-level reasoning before portfolio construction

Portfolio Judge

- Review all ticker decisions together
- Compare candidates jointly instead of independently
- Identify stronger and weaker ideas across the portfolio
- Convert stock-level outputs into a portfolio-level decision

Portfolio Weights

- Assign capital weights based on relative conviction
- Give larger weights to stronger opportunities
- Reduce or exclude weaker ideas
- Pass final weights to the risk management stage

Risk Management Layer

The final portfolio is not passed directly into backtesting. It is first checked by deterministic risk rules.

Rule-Based Safeguards

- Apply single-position caps
- Control sector exposure
- Enforce minimum holdings
- Normalize final allocations
- Redistribute weights when limits are violated
- Support defensive equal-weight fallback when needed

Position Limits

Prevent any one stock from becoming too large.

Sector Limits

Reduce concentration in the same sector.

Normalization

Ensure final portfolio weights sum correctly.

Fallback Logic

Keep the system operational when model-driven outputs are incomplete.

Hybrid Rule-Based + LLM Architecture

DebateTrader combines rule-based financial processing with LLM-based reasoning in a modular decision pipeline.

Rule-Based Components

- Data collection
- Feature building
- Risk constraints
- Backtesting logic

Why Rule-Based

- Transparent processing
- Stable portfolio safeguards
- Easier debugging and evaluation

Central Decision Pipeline

Data and Feature Layer

- Prices, fundamentals, macro data, news, and trends
- Standardized weekly feature snapshots
- Time-aligned inputs

Analyst Layer

- Technical Analyst
- Fundamental Analyst
- Macro Analyst
- News & Trend Analyst
- Structured analyst reports

Debate Layer

- Bull Agent
- Bear Agent
- Judge Agent
- Final ticker-level decision

Portfolio Layer

- Portfolio Judge
- Cross-ticker comparison
- Portfolio weight assignment

Risk and Evaluation Layer

- Deterministic constraints
- Weight normalization
- Historical backtesting

LLM Components

- Debate stage
- Portfolio judge
- Structured reasoning over evidence

Model Usage

- Debate: Groq-hosted Llama 3.1 8B Instant
- Portfolio stage: Claude Sonnet 4.6 via Anthropic
- Structured prompting with JSON-style outputs

Historical Backtesting Framework

Backtest Period
2025-08-04 to 2026-04-09

Horizon
36 weeks

Rebalancing
Weekly

Evaluation Mode
Paper Trading

Benchmarks

- Equal-weight six-stock portfolio
- SPY buy-and-hold
- 60/40 benchmark

Evaluation Metrics

- Total return
- Annualized return
- Sharpe ratio
- Max drawdown
- Win rate
- Average weekly return

The backtest evaluates DebateTrader over a fixed 36-week horizon using weekly rebalancing, standard portfolio benchmarks, and common performance metrics.

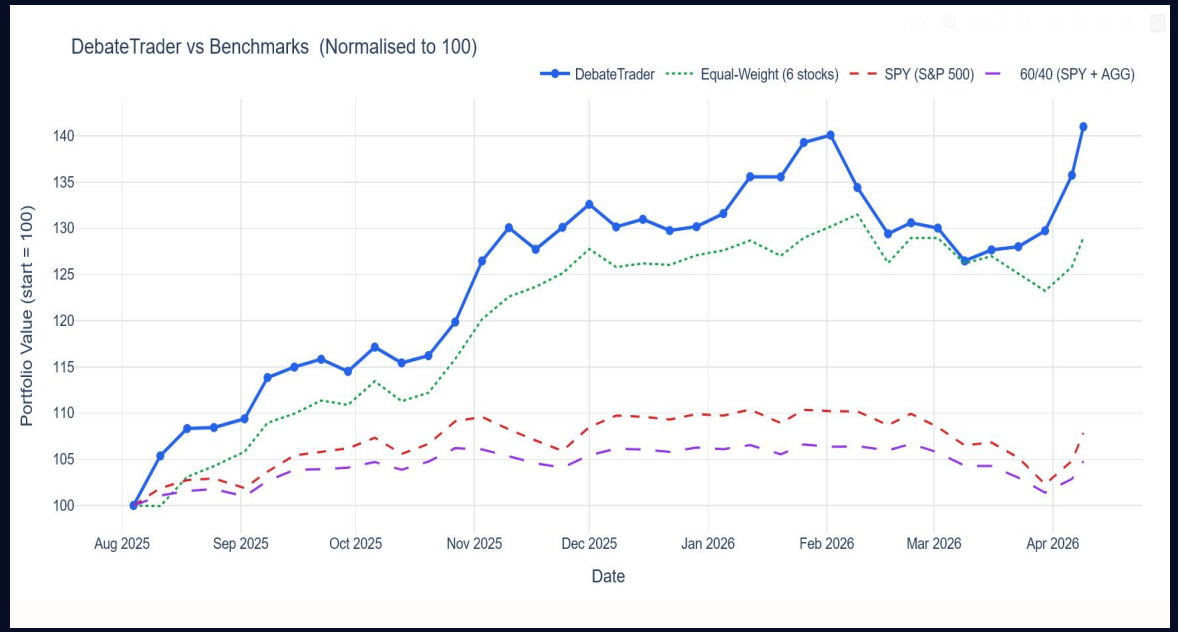
DebateTrader Achieved the Highest Raw Return

DebateTrader delivered the highest total return and annualized return across all compared strategies. However, its Sharpe ratio was slightly below the equal-weight benchmark, and its maximum drawdown was higher than all benchmarks. This suggests strong short-horizon return performance, but not uniform superiority on all risk-adjusted metrics.

Metric	DebateTrader	Equal-Weight	SPY B&H	60/40
Total return	+41.00%	+29.01%	+7.86%	+4.81%
Annualized return	+64.26%	+44.47%	+11.55%	+7.02%
Sharpe ratio (ann.)	3.0521	3.0670	1.1369	1.0892
Max drawdown	9.72%	6.29%	7.32%	4.94%
Win rate	72.2%	72.2%	55.6%	52.8%
Avg weekly return	+0.9850%	+0.7241%	+0.2198%	+0.1343%

Equity Curve Comparison

Normalized portfolio value for DebateTrader, Equal-Weight, SPY Buy-and-Hold, and 60/40 from 2025-08-04 to 2026-04-09



- DebateTrader finishes the period with the highest ending portfolio value.
- Performance leadership is not monotonic, and the strategy undergoes meaningful interim volatility.
- The equity curve complements the table by showing both cumulative return and the timing of drawdowns.

Key Findings

DebateTrader did not just generate stock picks; it produced a modular, inspectable, and dynamically differentiated portfolio process.

Modular Workflow

Structured intermediate outputs made the system easier to inspect, debug, and reuse across stages.

More Interpretable Reasoning

The Bull-Bear debate preserved competing arguments and made disagreement visible before a final decision was made.

Differentiated Allocation

The portfolio judge produced non-uniform allocations instead of collapsing into a simple equal-weight portfolio.

Consistent Portfolio Variation

Across 36 weekly outputs, the system produced 36 unique allocation patterns, with 3 to 6 non-zero holdings and an average of 4.53 holdings per week.

Responsible AI Considerations

The system improves interpretability and control, but its outputs should still be treated as experimental rather than deployment-ready investment advice.

Bias & Generalizability

- Evaluated on a small six-stock large-cap universe
- Results may not generalize to broader markets
- Performance may vary under different market regimes

Hallucination Controls

- Structured JSON outputs
- Rule-based feature extraction
- Evidence-constrained prompting
- Deterministic fallback logic

Privacy

- Uses public market, macro, news, and trend data
- External LLM calls still introduce governance concerns
- Stored reasoning traces require careful handling

Safety

- Research prototype only
- Used for paper trading and backtesting
- No direct broker execution
- Final allocations checked by rule-based risk controls

Demo

Live system walkthrough: Analyst reports → Bull-Bear debate → Portfolio allocation → Risk management → Backtest output

Link: https://hlj0908.github.io/debatetrader-demo-site/

Limitations and Future Work

DebateTrader shows promising results as a research prototype, but broader evaluation and more realistic deployment assumptions are needed before real-world use.

⚠️ Limitations

- Fixed six-stock universe limits generalizability
- Lightweight debate memory is still a simplified design
- External data coverage remains limited
- No dedicated factual-verification layer is included
- Backtesting does not include transaction costs, slippage, taxes, or live execution frictions

📈 Future Work

- Expand the stock universe beyond the current six-stock setup
- Evaluate debate quality more systematically
- Measure how debate changes final portfolio decisions
- Add transaction costs, tax assumptions, and more realistic portfolio constraints
- Improve factual verification and robustness of the pipeline

Conclusion

DebateTrader integrates structured evidence extraction, adversarial debate, portfolio-level judgment, rule-based risk safeguards, and historical backtesting.

Modular

- Separates data, analysis, debate, allocation, and evaluation
- Produces inspectable intermediate outputs

Interpretable

- Makes bullish and bearish reasoning explicit
- Preserves disagreement before final decisions

Research-Oriented

- Shows strong raw return in backtesting
- Still involves clear risk and scalability trade-offs